The role of creativity in generating options for open-ended value-based decisions

Seoyoung Lee[1,2], Akram Bakkour[1,2,3]

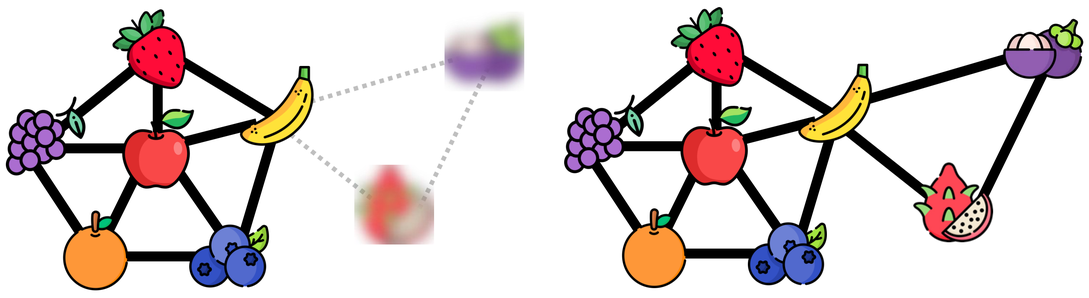
Department of Psychology[1], Institute for Mind and Biology[2], Neuroscience Institute[3], The University of Chicago

seoyounglee@uchicago.edu

Does creativity shape memory search in option generation and option evaluation in decision making?

Option generation requires memory

Open-ended value-based decision: What *fruit* would you eat as a snack?



Low creativity:
limited access to memory network

High creativity:
better access to memory network

Value-based decisions depend on access to memory



Low creativity:
less contextually appropriate decisions

High creativity:
more contextually appropriate decisions

Task design

N = 170

Imagination task

Written response to:

Imagine what your life will be like in the future (e.g., where you'll be, what you'll do, how you'll feel etc.)

Write as much detail as possible.

(5 minutes total)

Option generation task

Scenario: Imagine you are packing a snack for the day.

Question: *What kinds of fruits* would you pack as a snack?

Answer: apple, orange, strawberry, grape

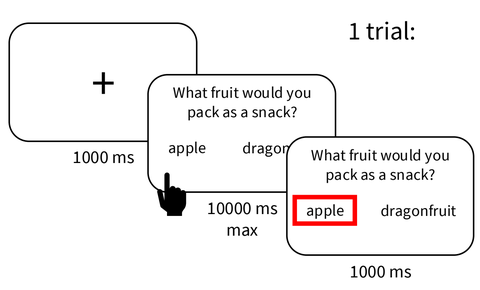
Question: *Which fruit* would you pack as a snack?

Answer: apple

(2 trials total)

Decision-making task

1 trial:



(Total number of trials depends on the number of options generated)

What fruit would you pack as a snack?

apple (self-generated)

dragonfruit (novel)

Answers from option generation task

Gemini

Prompt: *Generate words that are semantically similar and dissimilar to this set of words*

Semantically *similar*: pear, cherries

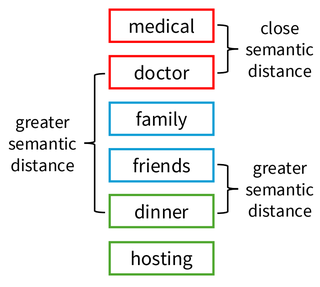
Semantically *dissimilar*: dragonfruit, fig

Divergent Semantic Integration (DSI) as a measure of creativity

Example of written response to imagination task:

"In 10 years, my life will probably be a lot busier. Right now, I am in medical school. 10 years from now, I will a full doctor. I am hoping to have a family and will probably be raising small children. I will be very busy and overwhelmed but hopefully very fulfilled, motivated, and happy. If I am preparing dinner for friends sometime in 10 years, I am imagining that I will have way more kitchen gadgets and funds to make a really nice dinner for a large group of people. I will be so happy to be hosting. Hopefully, the kids will have a babysitter so I am really able to focus on hosting. I will likely go to the grocery store and get a lot of really good ingredients to make something complex. I will also go to a HomeGoods store to get some themed decorations. I will think of a couple of nice table games to play with my friends. I will then get going on the cooking activities. Hopefully, I have some sort of help from my partner. I will prepare a really good meal using all of my kitchen gadgets. It will turn out wonderfully. All of my friends will be so grateful. It will be the best time."

Extent to which a text integrates divergent ideas by averaging the semantic distances between all words within the text

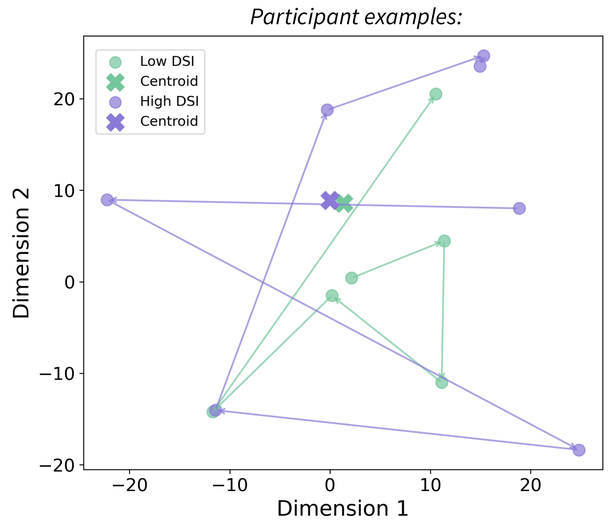


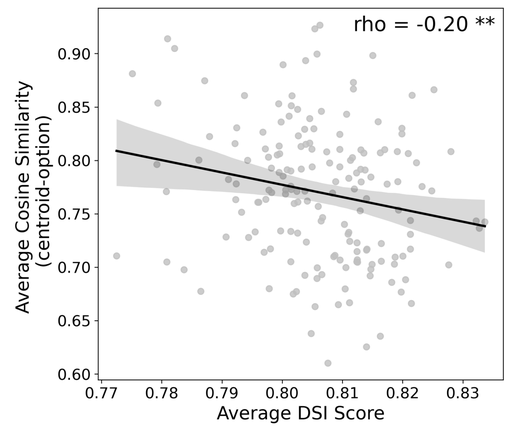
Higher DSI scores = response combines more semantically distant concepts reflecting more divergent and creative thinking

Higher creativity is associated with generating more semantically diverse options

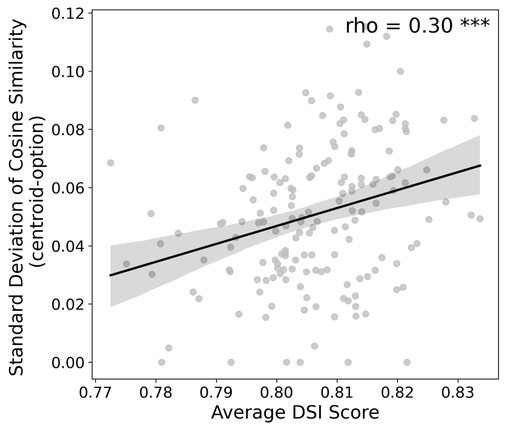
Centroid-Option

Participant examples:

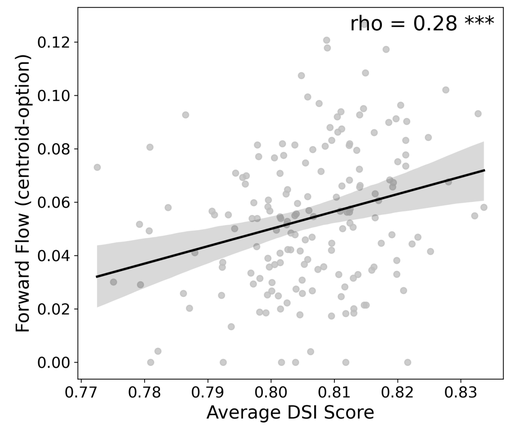




As creativity increases, participants generated options that were on average more semantically different from their typical response (I.e., centroid of their options)



As creativity increases, participants generated options that were more semantically variable from their typical response (i.e., options were more spread out)



As creativity increases, each successive option generated was more semantically different from an individual's typical response relative to all previously generated options

Higher creativity is associated with making more semantically appropriate decisions

Option generation task

Question: *What kinds of fruits* would you pack as a snack?

Answer: apple, orange, strawberry, grape

Cross-Encoder

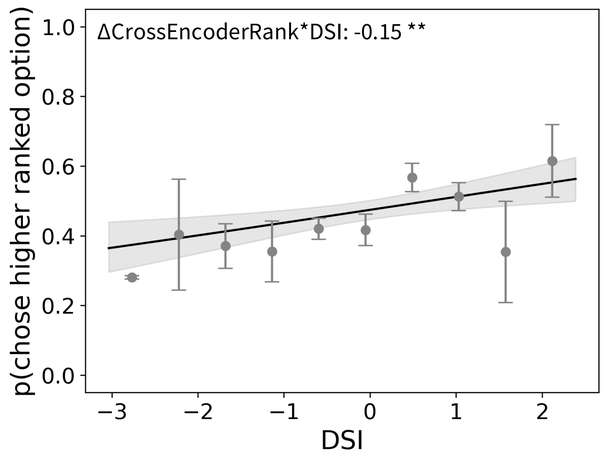
What kinds of fruits would you pack as a snack? → Bert

apple (repeat for every answer) → Bert

Bert → Classifier → Semantic similarity between question and answer [0, 1]

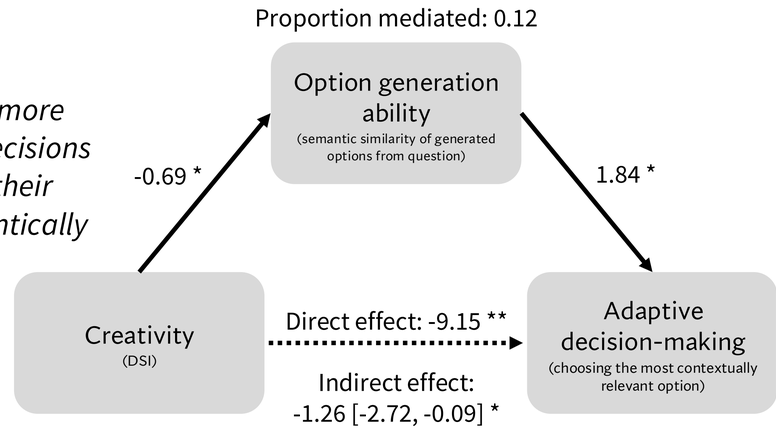
Ranking:
1. apple
2. strawberry
3. orange
4. grape

ΔCrossEncoderRank (for each decision) = right option rank – left option rank



As creativity increases, participants were more likely to choose the option that was more semantically similar to the question

More creative individuals make more adaptive (context-appropriate) decisions and this is partly explained by their tendency to generate more semantically diverse options



Conclusion

Creativity supports both broader semantic memory exploration in option generation and more adaptive (context-sensitive) decision-making